Class 14: RNASeq Mini-Project

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Background

The data for today's mini-project comes from a knock-down study of an important HOX gene.

Data Import

```
countData <- read.csv("GSE37704_featurecounts (2).csv", row.names = 1)
colData <-  read.csv("GSE37704_metadata.csv")
head(countData)
```

```
                 length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
                 SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

```r
colData
```

```
          id     condition
1 SRR493366 control_sirna
2 SRR493367 control_sirna
3 SRR493368 control_sirna
4 SRR493369      hoxa1_kd
5 SRR493370      hoxa1_kd
6 SRR493371      hoxa1_kd
```

Clean up (data tidying)

Q. Complete the code below to remove the troublesome first column from countData

```r
countData <- as.matrix(countData[,-1])
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
ENSG00000278566         0         0         0         0         0         0
ENSG00000273547         0         0         0         0         0         0
ENSG00000187634       124       123       205       207       212       258
```

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```r
countData = countData[-rowSums(countData!=0),]
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000188976      1637      1831      2383      1226      1326      1504
ENSG00000187961       120       153       180       236       255       357
ENSG00000187583        24        48        65        44        48        64
ENSG00000187642         4         9        16        14        16        16
ENSG00000188290        31        34        57       172       172       223
ENSG00000187608       244       289       386       373       422       430
```

DESeq Analysis

```r
library(DESeq2)
```

Setting up the DESeq object

```r
dds = DESeqDataSetFromMatrix(countData=countData,
                             colData=colData,
                             design=~condition)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```r
dds = DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

Running DESeq

```
dds
```

```
class: DESeqDataSet
dim: 19802 6
metadata(1): version
assays(4): counts mu H cooks
rownames(19802): ENSG00000188976 ENSG00000187961 ... ENSG00000277475
  ENSG00000268674
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(3): id condition sizeFactor
```

Getting results

```
resultsNames(dds)
```

```
[1] "Intercept"                           "condition_hoxa1_kd_vs_control_sirna"
```

```
res = results(dds)
res
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 19802 rows and 6 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000188976 1651.1671     -0.6926547 0.0548586 -12.626185 1.51443e-36
ENSG00000187961  209.6376      0.7298186 0.1318816   5.533895 3.13197e-08
ENSG00000187583   47.2548      0.0406457 0.2719379   0.149467 8.81185e-01
ENSG00000187642   11.9797      0.5428889 0.5216403   1.040734 2.97999e-01
ENSG00000188290  108.9228      2.0571342 0.1969309  10.445970 1.52885e-25
...                   ...            ...       ...        ...         ...
ENSG00000277856    0.000             NA        NA         NA          NA
ENSG00000275063    0.000             NA        NA         NA          NA
ENSG00000271254  181.594      -0.609603  0.141347   -4.31283 1.61178e-05
ENSG00000277475    0.000             NA        NA         NA          NA
```

```
ENSG00000268674     0.000             NA         NA          NA           NA
                       padj
                  <numeric>
ENSG00000188976 1.81682e-35
ENSG00000187961 1.11424e-07
ENSG00000187583 9.16701e-01
ENSG00000187642 3.97259e-01
ENSG00000188290 1.29290e-24
...                      ...
ENSG00000277856          NA
ENSG00000275063          NA
ENSG00000271254 4.47015e-05
ENSG00000277475          NA
ENSG00000268674          NA
```

> Q. Call the summary() function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```r
summary(res)
```

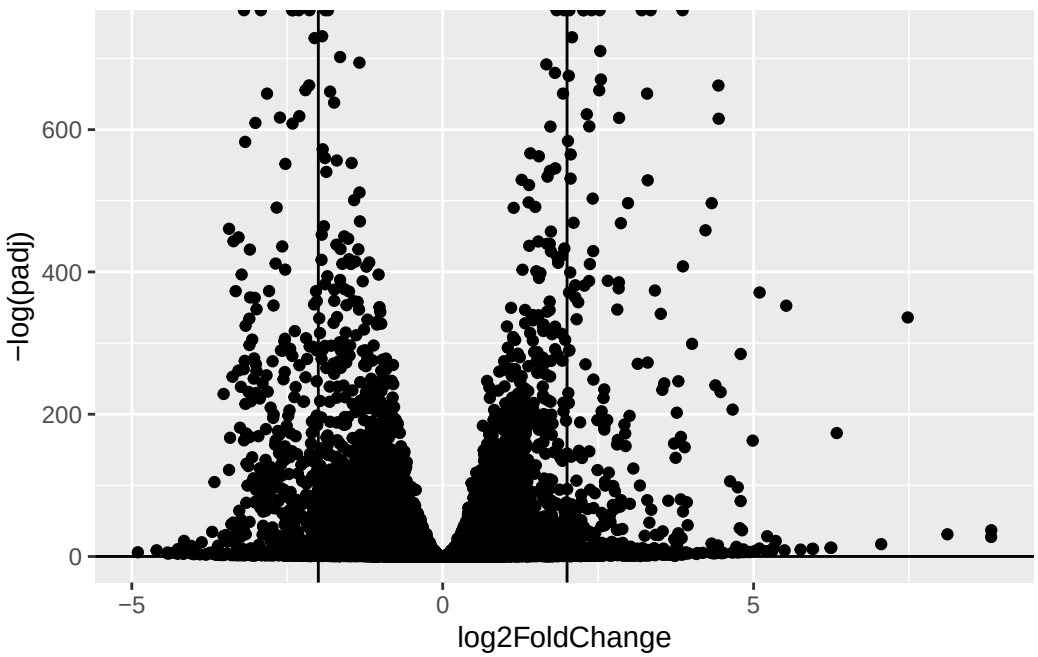
```
out of 15973 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 4357, 27%
LFC < 0 (down)     : 4408, 28%
outliers [1]       : 0, 0%
low counts [2]     : 1529, 9.6%
(mean count < 1)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Volcano Plot

Now we will make a volcano plot, a commonly produced visualization from this type of data that we introduced last day. Basically it's a plot of log2 fold change vs -log adjusted p-value.

```r
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point() + geom_vline(xintercept = c(-2,+2)) +geom_hline(yintercept = 0.05)
```

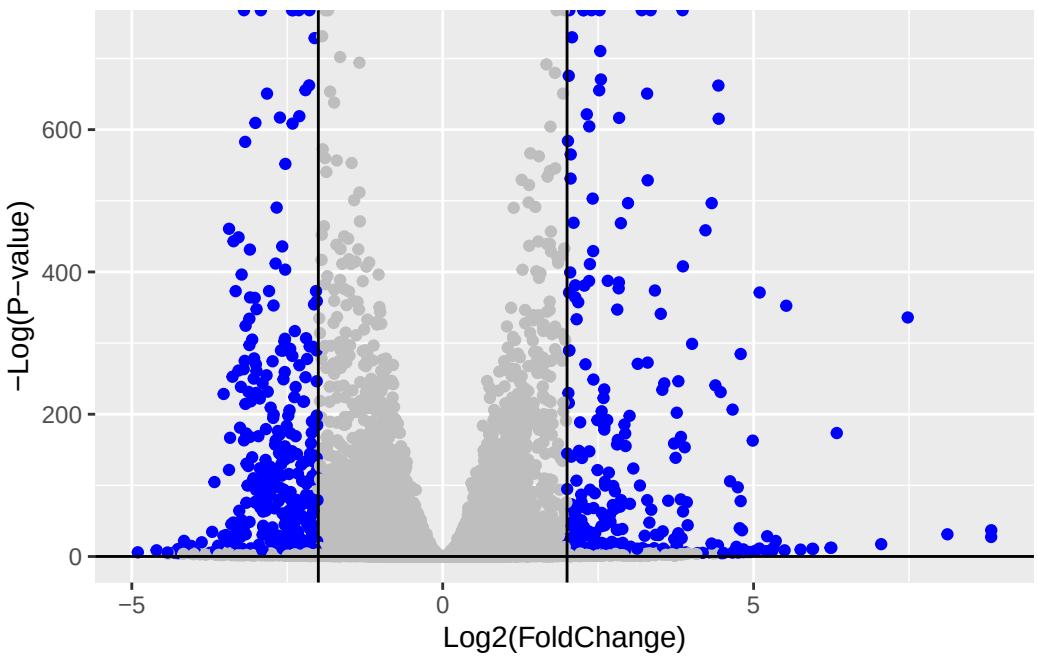
```
Warning: Removed 5358 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
mycols <- rep("gray", nrow(res) )


mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"

mycols[ res$padj > 0.01 ] <- "gray"

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(col =mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept  = c(-2,2)) +
  geom_hline(yintercept = 0.05)
```

```
Warning: Removed 5358 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Add Annotation

Since we mapped and counted against the Ensembl annotation, our results only have information about Ensembl gene IDs. However, our pathway analysis downstream will use KEGG pathways, and genes in KEGG pathways are annotated with Entrez gene IDs. So lets add them as we did the last day.

> Q.Use the mapIDs() function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```
res$symbol = mapIds(org.Hs.eg.db,
                    keys= rownames(res),
                    keytype="ENSEMBL",
                    column="SYMBOL",
                    multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$entrez = mapIds(org.Hs.eg.db,
                    keys=rownames(res),
                    keytype="ENSEMBL",
                    column="ENTREZID",
                    multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$name =   mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="ENTREZID",
                    multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
head(res, 10)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 10 rows and 9 columns
                   baseMean log2FoldChange     lfcSE       stat      pvalue
                  <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000188976 1651.167084     -0.6926547 0.0548586 -12.626185 1.51443e-36
ENSG00000187961  209.637584      0.7298186 0.1318816   5.533895 3.13197e-08
```

```
ENSG00000187583   47.254807      0.0406457 0.2719379   0.149467 8.81185e-01
ENSG00000187642   11.979708      0.5428889 0.5216403   1.040734 2.97999e-01
ENSG00000188290  108.922825      2.0571342 0.1969309  10.445970 1.52885e-25
ENSG00000187608  350.714974      0.2574511 0.1027442   2.505747 1.22193e-02
ENSG00000188157 9128.396655      0.3899749 0.0467229   8.346549 7.02863e-17
ENSG00000237330    0.158192      0.7860197 4.0804729   0.192630 8.47249e-01
ENSG00000131591  156.478099      0.1966549 0.1456338   1.350339 1.76907e-01
ENSG00000162571    0.000000             NA        NA         NA           NA
                     padj      symbol      entrez        name
                <numeric> <character> <character> <character>
ENSG00000188976 1.81682e-35       NOC2L       26155       26155
ENSG00000187961 1.11424e-07      KLHL17      339451      339451
ENSG00000187583 9.16701e-01     PLEKHN1       84069       84069
ENSG00000187642 3.97259e-01       PERM1       84808       84808
ENSG00000188290 1.29290e-24        HES4       57801       57801
ENSG00000187608 2.32629e-02       ISG15        9636        9636
ENSG00000188157 4.12689e-16        AGRN      375790      375790
ENSG00000237330          NA      RNF223      401934      401934
ENSG00000131591 2.56551e-01    C1orf159       54991       54991
ENSG00000162571          NA      TTLL10      254173      254173
```

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res = res[order(res$pvalue),]
write.csv(res, file="deseq_results.csv")
```

Pathway Analysis

Here we are going to use the gage package for pathway analysis. Once we have a list of enriched pathways, we're going to use the pathview package to draw pathway diagrams, shading the molecules in the pathway by their degree of up/down-regulation.

KEGG

First let's load the packages and setup the KEGG data-sets we need

```
library(pathview)
```

```
##############################################################################
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at http://www.kegg.jp/kegg/legal.html).
##############################################################################
```

```r
library(gage)
```

```r
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"

$`hsa00230 Purine metabolism`
  [1] "100"    "10201"  "10606"  "10621"  "10622"  "10623"  "107"    "10714"
  [9] "108"    "10846"  "109"    "111"    "11128"  "11164"  "112"    "113"
```

```
 [17] "114"    "115"    "122481" "122622" "124583" "132"    "158"    "159"
 [25] "1633"   "171568" "1716"   "196883" "203"    "204"    "205"    "221823"
 [33] "2272"   "22978"  "23649"  "246721" "25885"  "2618"   "26289"  "270"
 [41] "271"    "27115"  "272"    "2766"   "2977"   "2982"   "2983"   "2984"
 [49] "2986"   "2987"   "29922"  "3000"   "30833"  "30834"  "318"    "3251"
 [57] "353"    "3614"   "3615"   "3704"   "377841" "471"    "4830"   "4831"
 [65] "4832"   "4833"   "4860"   "4881"   "4882"   "4907"   "50484"  "50940"
 [73] "51082"  "51251"  "51292"  "5136"   "5137"   "5138"   "5139"   "5140"
 [81] "5141"   "5142"   "5143"   "5144"   "5145"   "5146"   "5147"   "5148"
 [89] "5149"   "5150"   "5151"   "5152"   "5153"   "5158"   "5167"   "5169"
 [97] "51728"  "5198"   "5236"   "5313"   "5315"   "53343"  "54107"  "5422"
[105] "5424"   "5425"   "5426"   "5427"   "5430"   "5431"   "5432"   "5433"
[113] "5434"   "5435"   "5436"   "5437"   "5438"   "5439"   "5440"   "5441"
[121] "5471"   "548644" "55276"  "5557"   "5558"   "55703"  "55811"  "55821"
[129] "5631"   "5634"   "56655"  "56953"  "56985"  "57804"  "58497"  "6240"
[137] "6241"   "64425"  "646625" "654364" "661"    "7498"   "8382"   "84172"
[145] "84265"  "84284"  "84618"  "8622"   "8654"   "87178"  "8833"   "9060"
[153] "9061"   "93034"  "953"    "9533"   "954"    "955"    "956"    "957"
[161] "9583"   "9615"
```

The main `gage()` function requires a named vector of fold changes, where the names of the values are the Entrez gene IDs.

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
     1266     54855      1465      2034      2150      6659
-2.422656  3.202017 -2.313674 -1.887955  3.344573  2.392353
```

Now time to run the gage pathway analysis and look at the objects returned from gage()

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

Now let's look at the first few pathway results

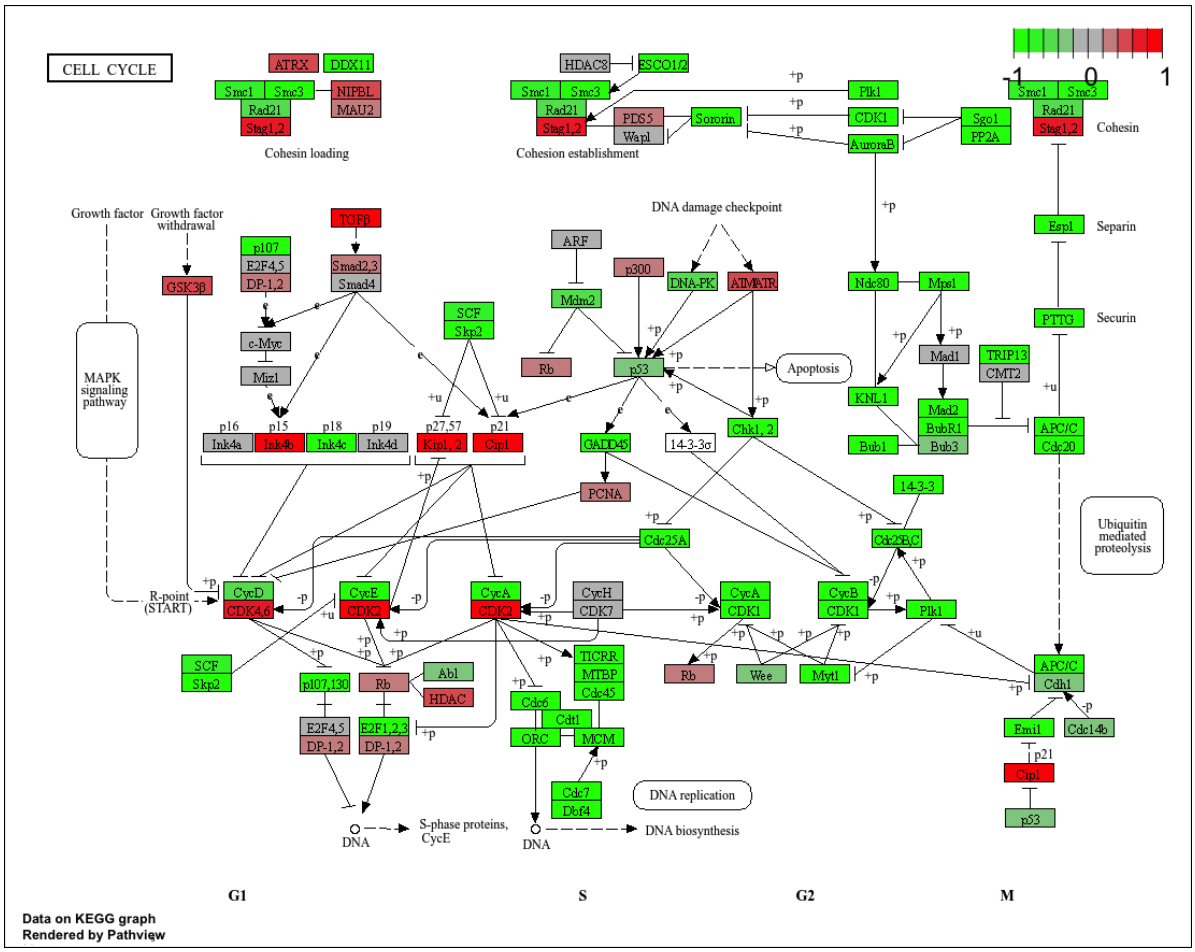
```r
head(keggres$less)
```

```
                                       p.geomean stat.mean        p.val
hsa04110 Cell cycle                   7.088745e-06 -4.432237 7.088745e-06
hsa03030 DNA replication              9.433633e-05 -3.951509 9.433633e-05
hsa03013 RNA transport                1.162033e-03 -3.080122 1.162033e-03
hsa04114 Oocyte meiosis               2.566266e-03 -2.826979 2.566266e-03
hsa03440 Homologous recombination     3.068576e-03 -2.852682 3.068576e-03
hsa00010 Glycolysis / Gluconeogenesis 4.363460e-03 -2.663551 4.363460e-03
                                            q.val set.size         exp1
hsa04110 Cell cycle                   0.001162554      124 7.088745e-06
hsa03030 DNA replication              0.007735579       36 9.433633e-05
hsa03013 RNA transport                0.063524478      149 1.162033e-03
hsa04114 Oocyte meiosis               0.100649277      112 2.566266e-03
hsa03440 Homologous recombination     0.100649277       28 3.068576e-03
hsa00010 Glycolysis / Gluconeogenesis 0.119267900       65 4.363460e-03
```

Now, let's try out the pathview() function to make a pathway plot with our RNA-Seq expression results shown in color. First we will look at the genes related to the cell cycle.

```r
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14

Info: Writing image file hsa04110.pathview.png
```



Now, let's process our results more automatically of the top 5 upregulated pathways and then process to get the IDs needed for the `pathview()` function.

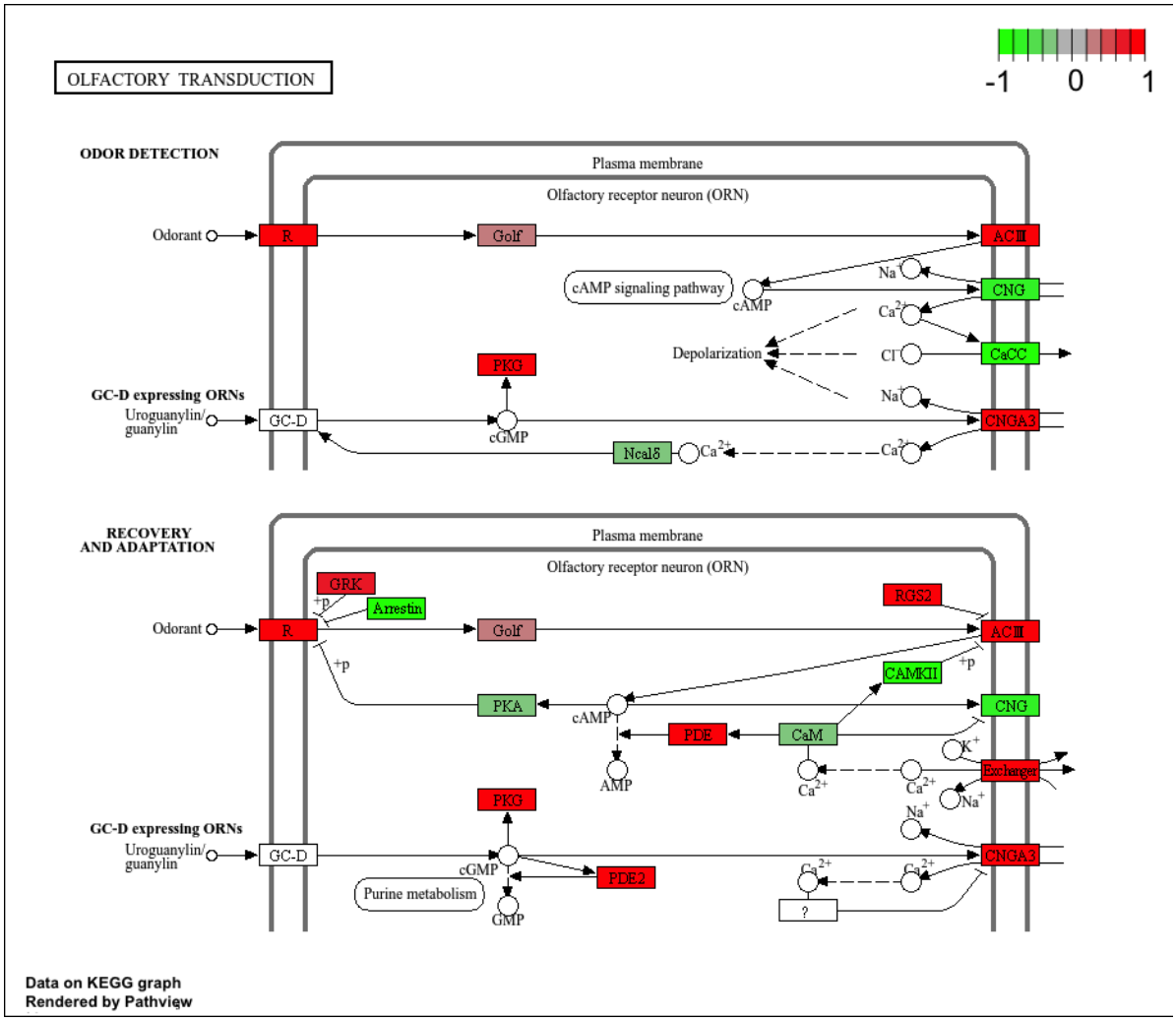
```
keggrespathways <- rownames(keggres$greater)[1:5]

# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

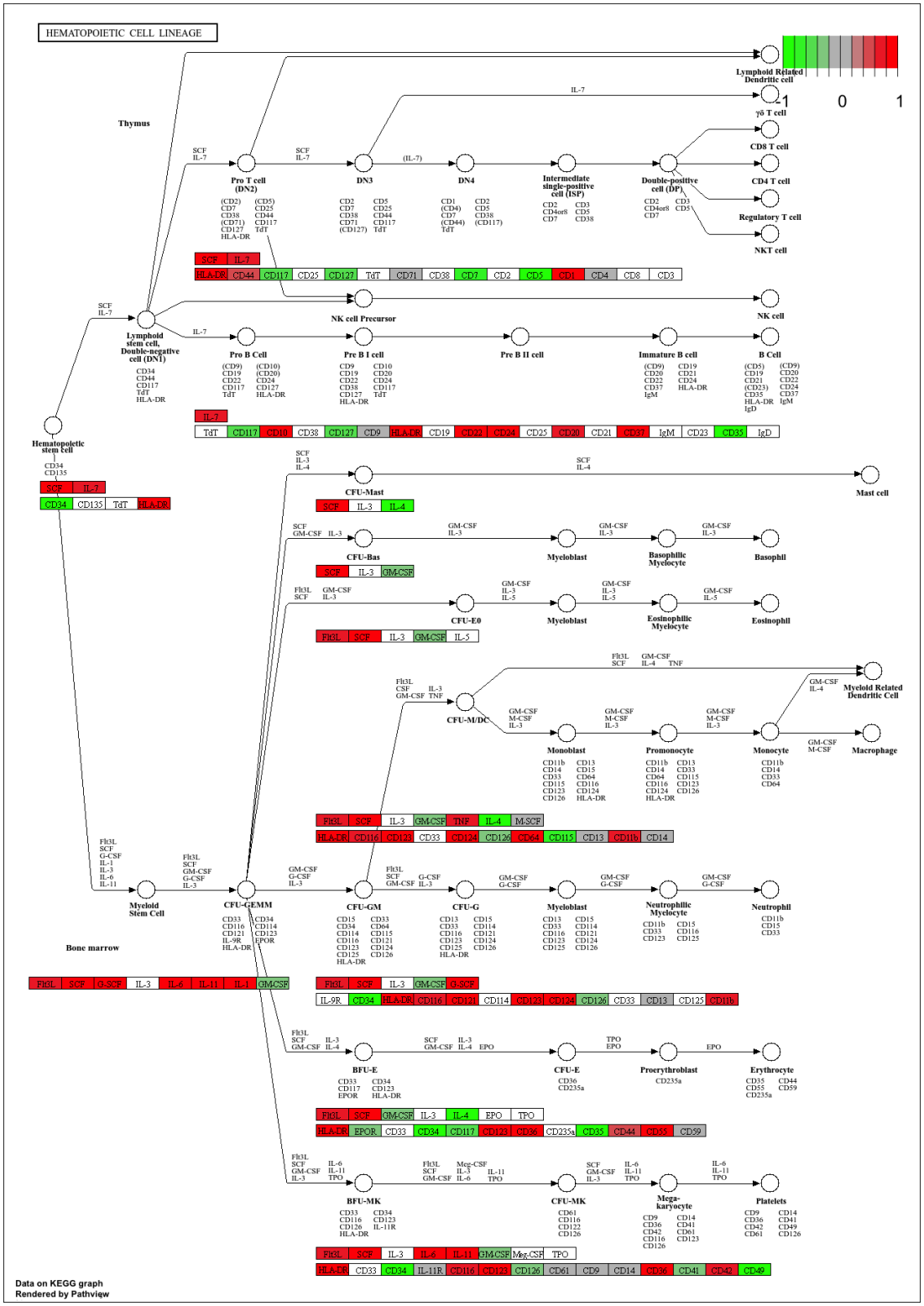
```
[1] "hsa04740" "hsa04640" "hsa00140" "hsa04630" "hsa04976"
```

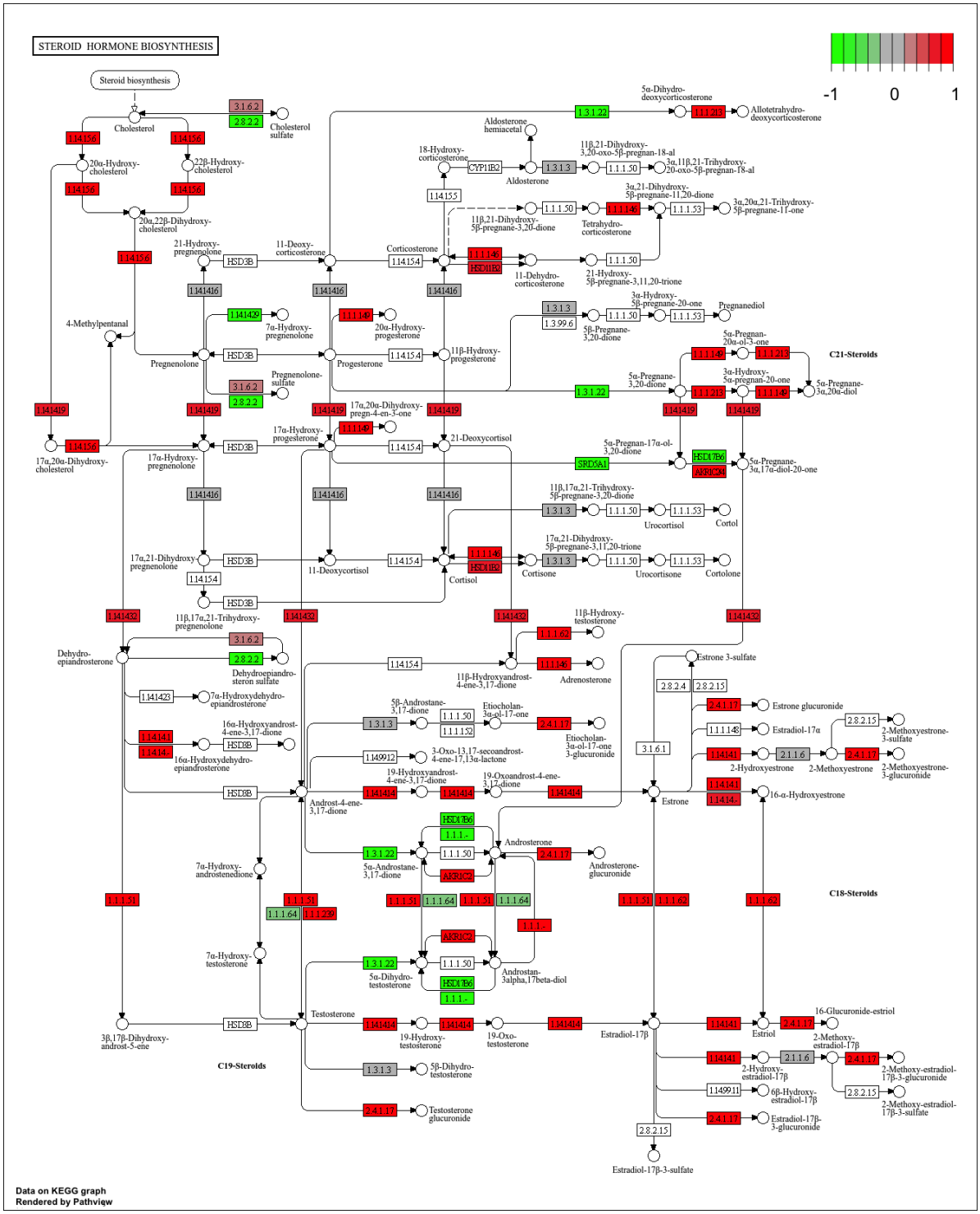
Finally, let us pas these IDs into the `pathview()` function to make plots for the top 5 pathways.

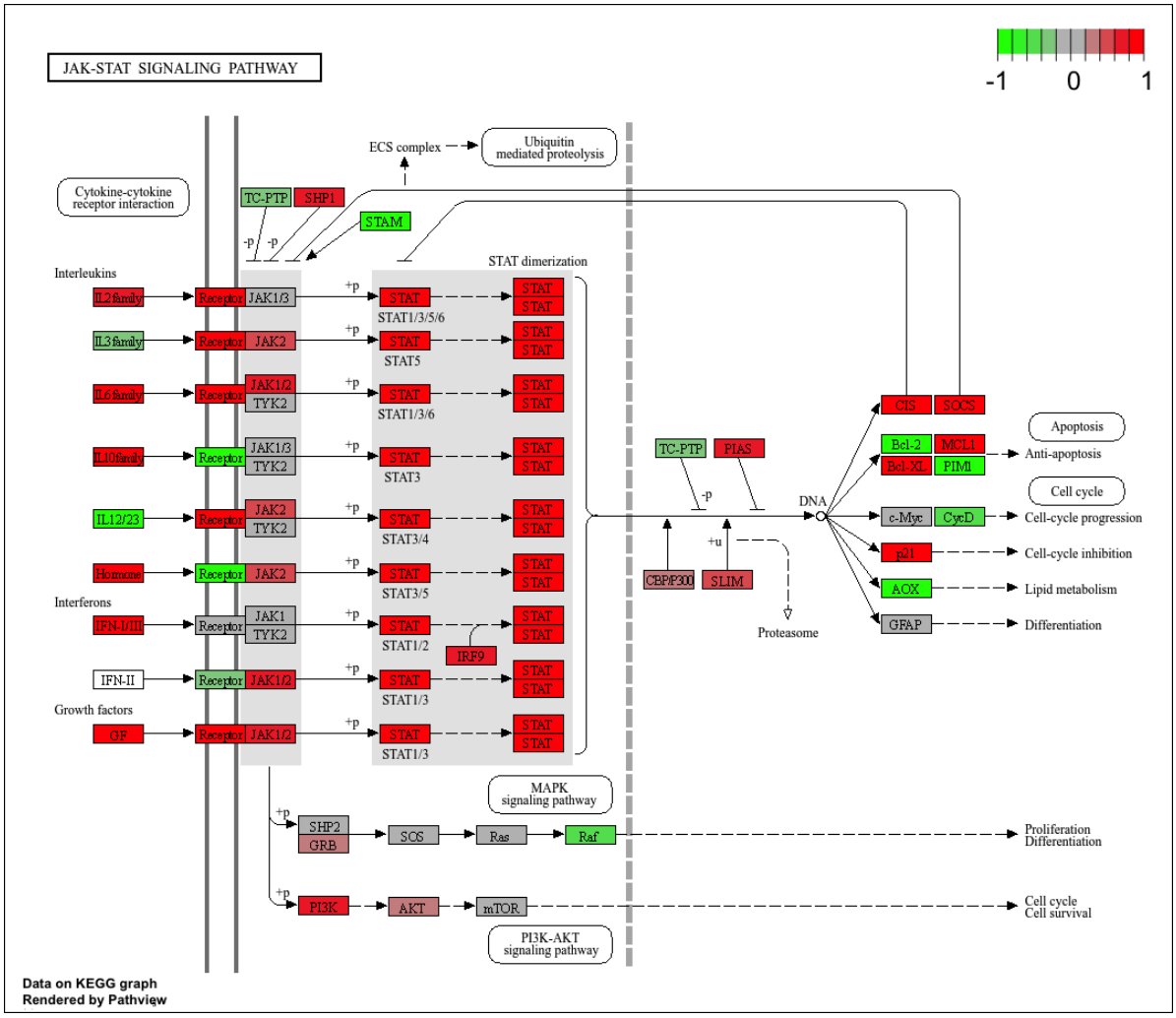
```r
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

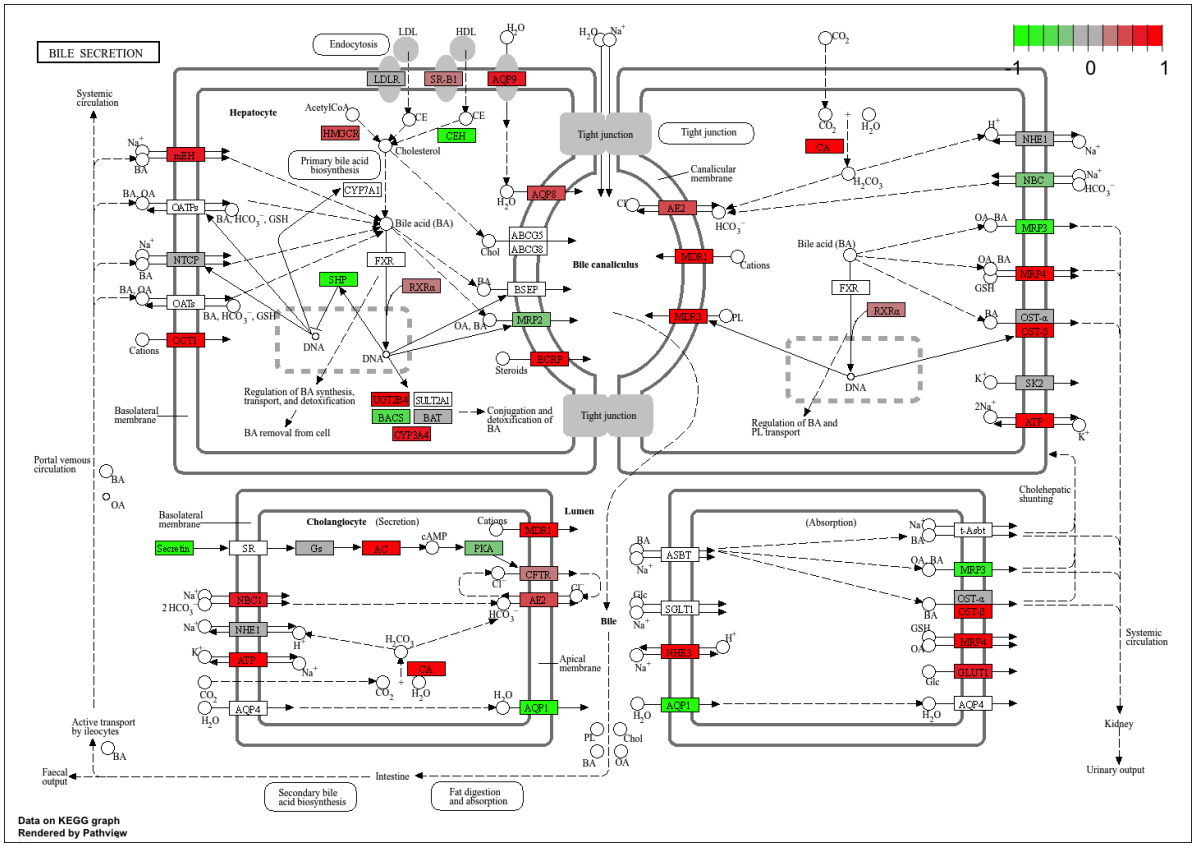
```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14

Info: Writing image file hsa04740.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14

Info: Writing image file hsa04976.pathview.png
```


OLFACTORY TRANSDUCTION
-1
0
1
ODOR DETECTION
Plasma membrane
Olfactory receptor neuron (ORN)
Odorant
R
Golf
ACIII
cAMP signaling pathway
cAMP
CNG
Depolarization
CaCC
PKG
GC-D expressing ORNs
Uroguanylin/
guanylin
GC-D
cGMP
Ncal8
CNGA3
RECOVERY
AND ADAPTATION
GRK
Arrestin
RGS2
CAMKII
PKA
PDE
CaM
Exchanger
AMP
PDE2
Purine metabolism
GMP
?
Data on KEGG graph
Rendered by Pathview


HEMATOPOIETIC CELL LINEAGE
Lymphoid Related Dendritic cell
γδ T cell
CD8 T cell
CD4 T cell
Regulatory T cell
NKT cell
Thymus
Pro T cell (DN2)
DN3
DN4
Intermediate single-positive cell (ISP)
Double-positive cell (DP)
NK cell Precursor
NK cell
Lymphoid stem cell, Double-negative cell (DN1)
Pro B Cell
Pre B I cell
Pre B II cell
Immature B cell
B Cell
Hematopoietic stem cell
CFU-Mast
Mast cell
CFU-Bas
Myeloblast
Basophilic Myelocyte
Basophil
CFU-E0
Eosinophilic Myelocyte
Eosinophil
CFU-M/DC
Myeloid Related Dendritic Cell
Monoblast
Promonocyte
Monocyte
Macrophage
Myeloid Stem Cell
CFU-GEMM
CFU-GM
CFU-G
Neutrophilic Myelocyte
Neutrophil
Bone marrow
BFU-E
CFU-E
Proerythroblast
Erythrocyte
BFU-MK
CFU-MK
Mega-karyocyte
Platelets
Data on KEGG graph
Rendered by Pathview


STEROID HORMONE BIOSYNTHESIS
-1
0
1
Steroid biosynthesis
Cholesterol
Cholesterol sulfate
20α-Hydroxy-cholesterol
22β-Hydroxy-cholesterol
20α,22β-Dihydroxy-cholesterol
21-Hydroxy-pregnenolone
11-Deoxy-corticosterone
Corticosterone
11-Dehydro-corticosterone
21-Hydroxy-5β-pregnane-3,11,20-trione
Aldosterone hemiacetal
Aldosterone
18-Hydroxy-corticosterone
CYP11B2
11β,21-Dihydroxy-3,20-oxo-5β-pregnan-18-al
3α,11β,21-Trihydroxy-20-oxo-5β-pregnan-18-al
5α-Dihydro-deoxycorticosterone
Allotetrahydro-deoxycorticosterone
11β,21-Dihydroxy-5β-pregnane-3,20-dione
Tetrahydro-corticosterone
3α,21-Dihydroxy-5β-pregnane-11,20-dione
3α,20α,21-Trihydroxy-5β-pregnane-11-one
4-Methylpentanal
Pregnenolone
7α-Hydroxy-pregnenolone
20α-Hydroxy-progesterone
Progesterone
11β-Hydroxy-progesterone
Pregnenolone-sulfate
3α-Hydroxy-5β-pregnane-20-one
Pregnanediol
5β-Pregnane-3,20-dione
5α-Pregnan-20α-ol-3-one
5α-Pregnane-3,20-dione
3α-Hydroxy-5α-pregnan-20-one
5α-Pregnane-3α,20α-diol
C21-Sterolds
17α,20α-Dihydroxy-pregn-4-en-3-one
17α,20α-Dihydroxy-cholesterol
17α-Hydroxy-pregnenolone
17α-Hydroxy-progesterone
21-Deoxycortisol
5α-Pregnan-17α-ol-3,20-dione
5α-Pregnane-3α,17α-diol-20-one
11β,17α,21-Trihydroxy-5β-pregnane-3,20-dione
Urocortisol
Cortol
17α,21-Dihydroxy-pregnenolone
11-Deoxycortisol
Cortisol
17α,21-Dihydroxy-5β-pregnane-3,11,20-trione
Cortisone
Urocortisone
Cortolone
11β,17α,21-Trihydroxy-pregnenolone
Dehydro-epiandrosterone
Dehydroepiandro-steron sulfate
7α-Hydroxydehydro-epiandrosterone
16α-Hydroxyandrost-4-ene-3,17-dione
16α-Hydroxydehydro-epiandrosterone
11β-Hydroxy-testosterone
11β-Hydroxyandrost-4-ene-3,17-dione
Adrenosterone
Estrone 3-sulfate
5β-Androstane-3,17-dione
Etiocholan-3α-ol-17-one
Etiocholan-3α-ol-17-one 3-glucuronide
3-Oxo-13,17-secoandrost-4-ene-17,13α-lactone
19-Hydroxyandrost-4-ene-3,17-dione
19-Oxoandrost-4-ene-3,17-dione
Androst-4-ene-3,17-dione
Estrone
Estrone glucuronide
Estradiol-17α
2-Methoxyestrone-3-sulfate
2-Hydroxyestrone
2-Methoxyestrone
2-Methoxyestrone-3-glucuronide
16-α-Hydroxyestrone
Androsterone
5α-Androstane-3,17-dione
Androsterone-glucuronide
7α-Hydroxy-androstenedione
C18-Sterolds
7α-Hydroxy-testosterone
5α-Dihydro-testosterone
Androstan-3alpha,17beta-diol
Testosterone
3β,17β-Dihydroxy-androst-5-ene
19-Hydroxy-testosterone
19-Oxo-testosterone
Estradiol-17β
16-Glucuronide-estriol
Estriol
2-Methoxy-estradiol-17β
2-Hydroxy-estradiol-17β
2-Methoxy-estradiol-17β-3-glucuronide
6β-Hydroxy-estradiol-17β
2-Methoxy-estradiol-17β-3-sulfate
Estradiol-17β-3-glucuronide
C19-Sterolds
5β-Dihydro-testosterone
Testosterone glucuronide
Estradiol-17β-3-sulfate
Data on KEGG graph
Rendered by Pathview


JAK-STAT SIGNALING PATHWAY
-1
0
1
ECS complex
Ubiquitin mediated proteolysis
Cytokine-cytokine receptor interaction
TC-PTP
SHP1
STAM
-p
-p
STAT dimerization
Interleukins
IL2family
IL3family
IL6family
IL10family
IL12/23
Hormone
Interferons
IFN-I/III
IFN-II
Growth factors
GF
Receptor
JAK1/3
JAK2
JAK1/2
TYK2
JAK1/3
TYK2
JAK2
TYK2
JAK2
JAK1
TYK2
JAK1/2
JAK1/2
+p
STAT
STAT1/3/5/6
STAT5
STAT1/3/6
STAT3
STAT3/4
STAT3/5
STAT1/2
STAT1/3
STAT1/3
IRF9
TC-PTP
PIAS
-p
+u
CBP/P300
SLIM
Proteasome
DNA
CIS
SOCS
Bcl-2
MCL1
Bcl-XL
PIM1
c-Myc
CycD
p21
AOX
GFAP
Apoptosis
Anti-apoptosis
Cell cycle
Cell-cycle progression
Cell-cycle inhibition
Lipid metabolism
Differentiation
MAPK signaling pathway
SHP2
GRB
SOS
Ras
Raf
Proliferation
Differentiation
PI3K
AKT
mTOR
Cell cycle
Cell survival
PI3K-AKT signaling pathway
Data on KEGG graph
Rendered by Pathview



Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

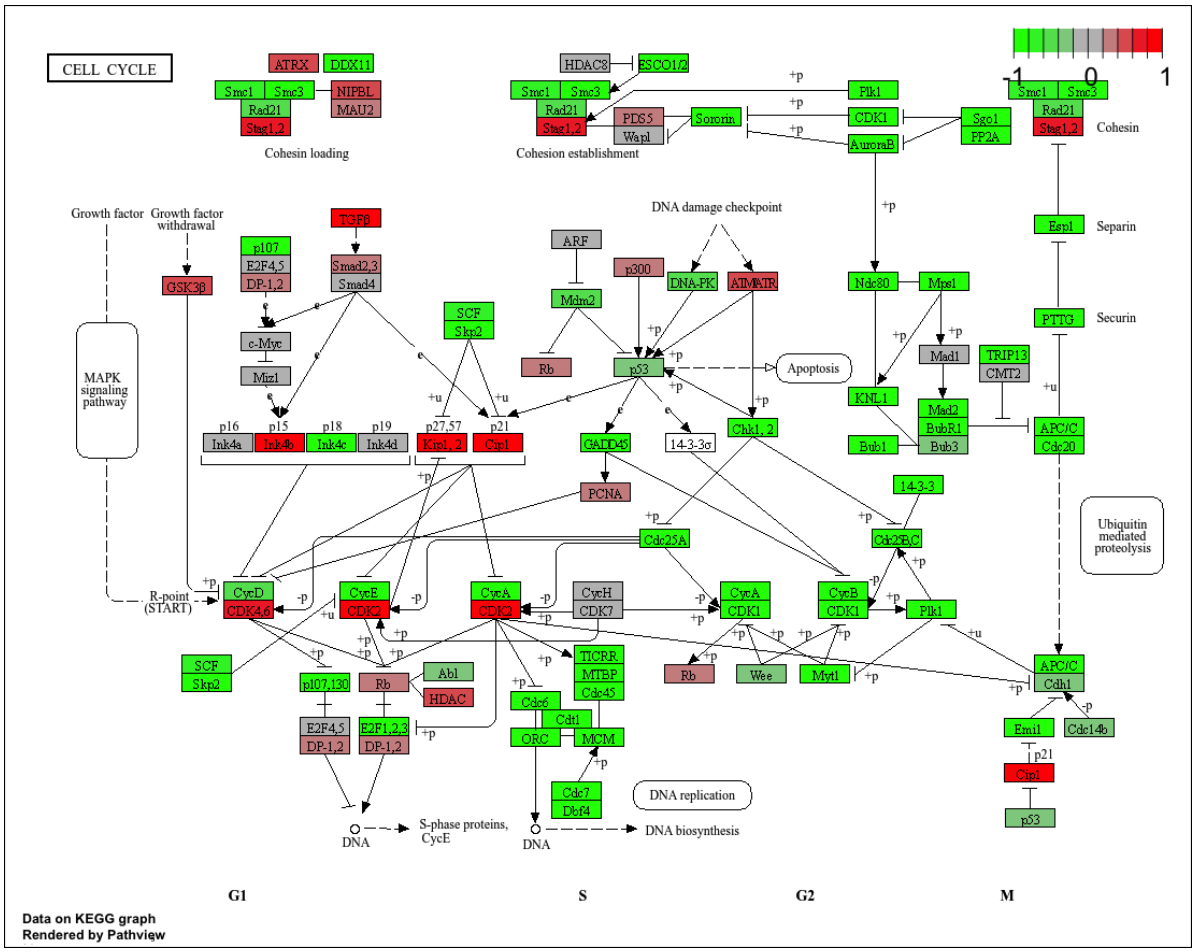
```
keggrespathways <- rownames(keggres$less)[1:5]

# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

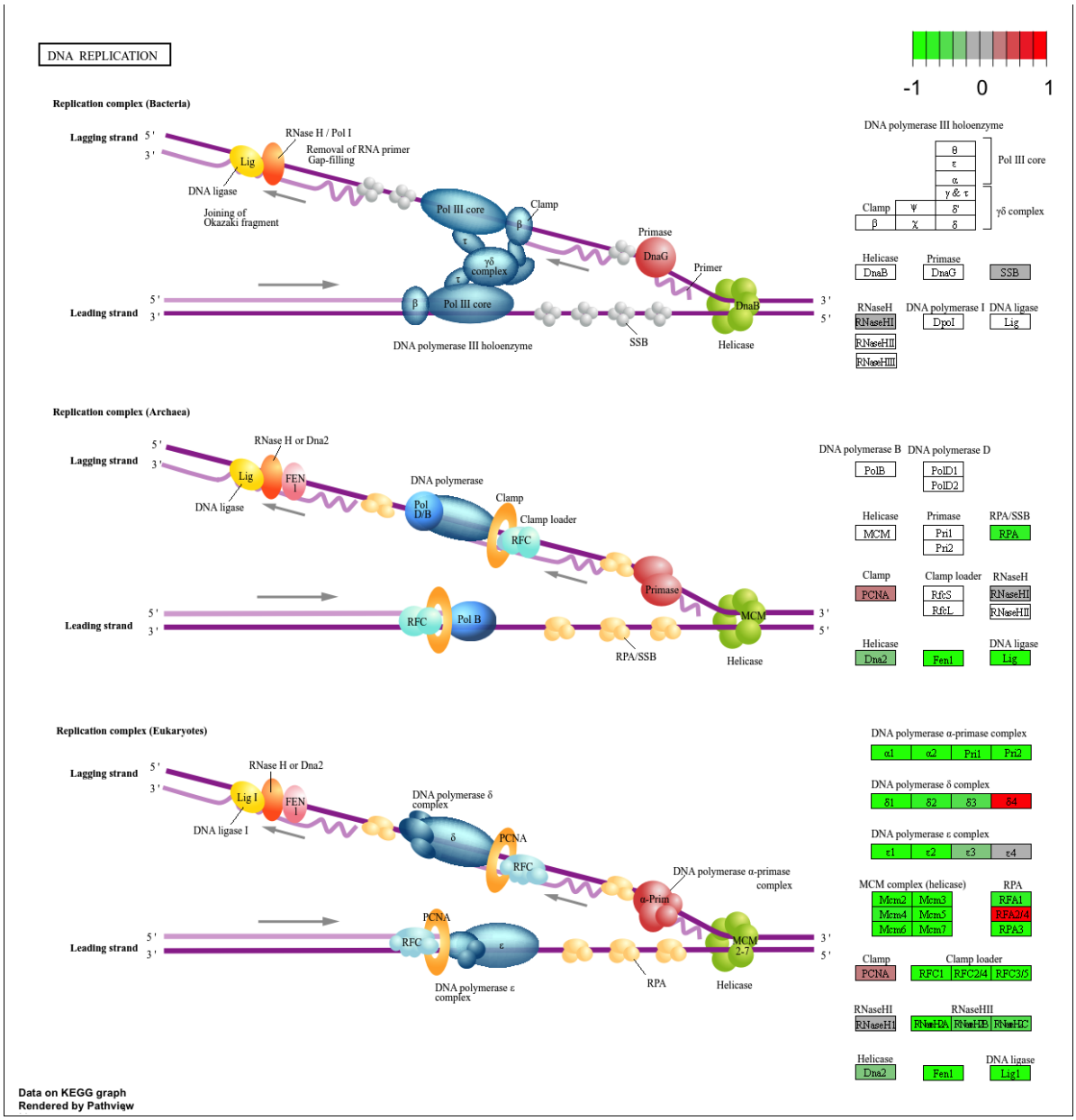
```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa04114" "hsa03440"
```

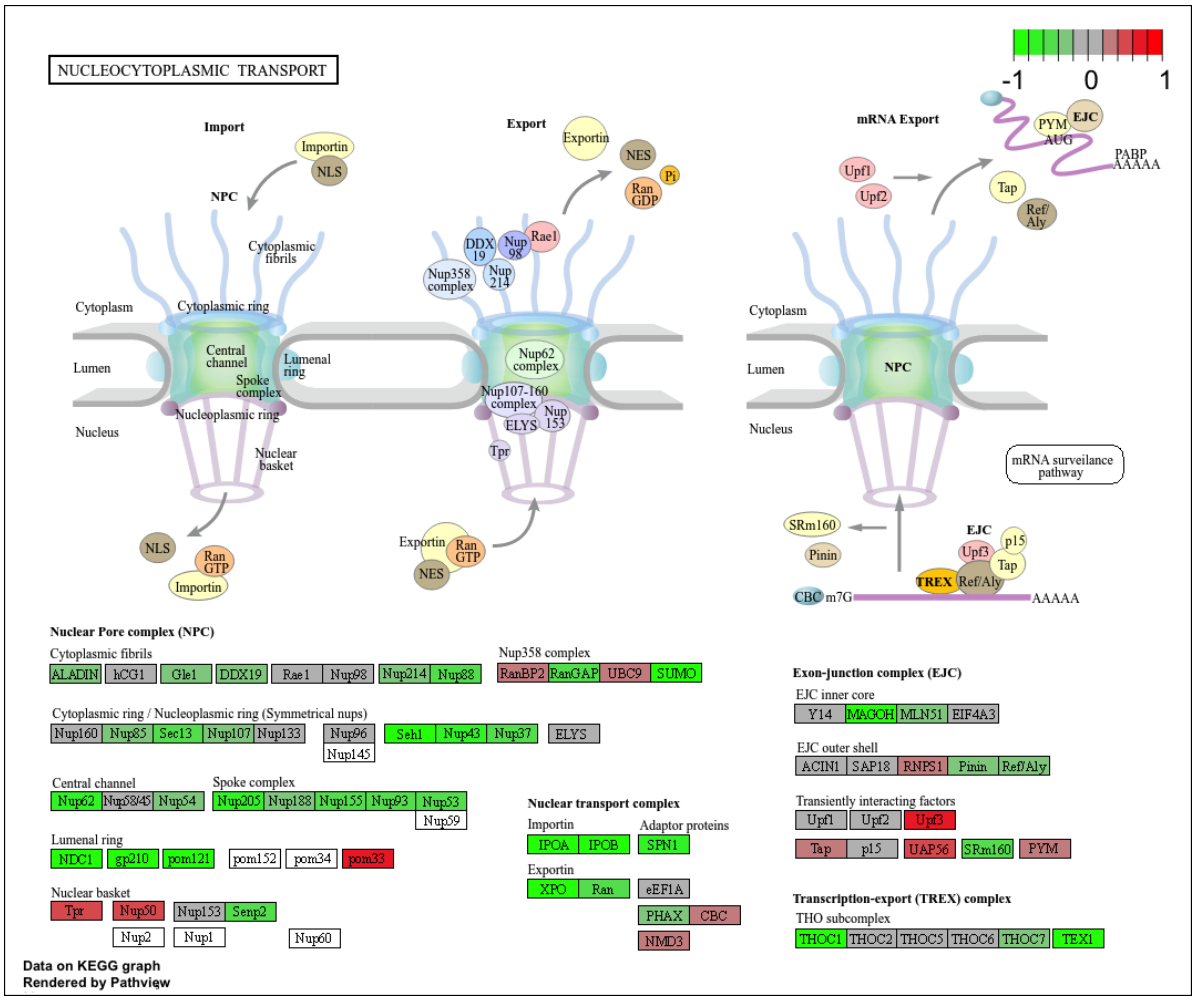
```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

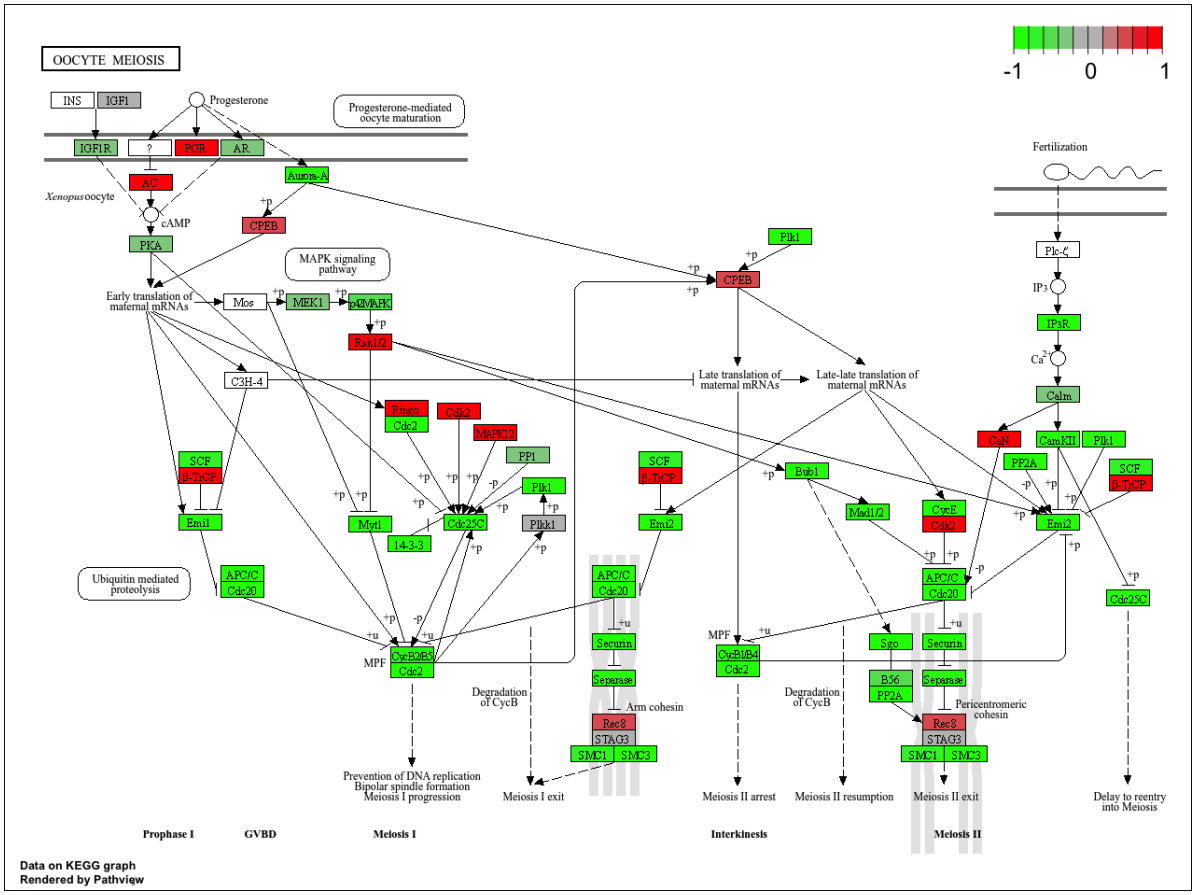
```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14
```

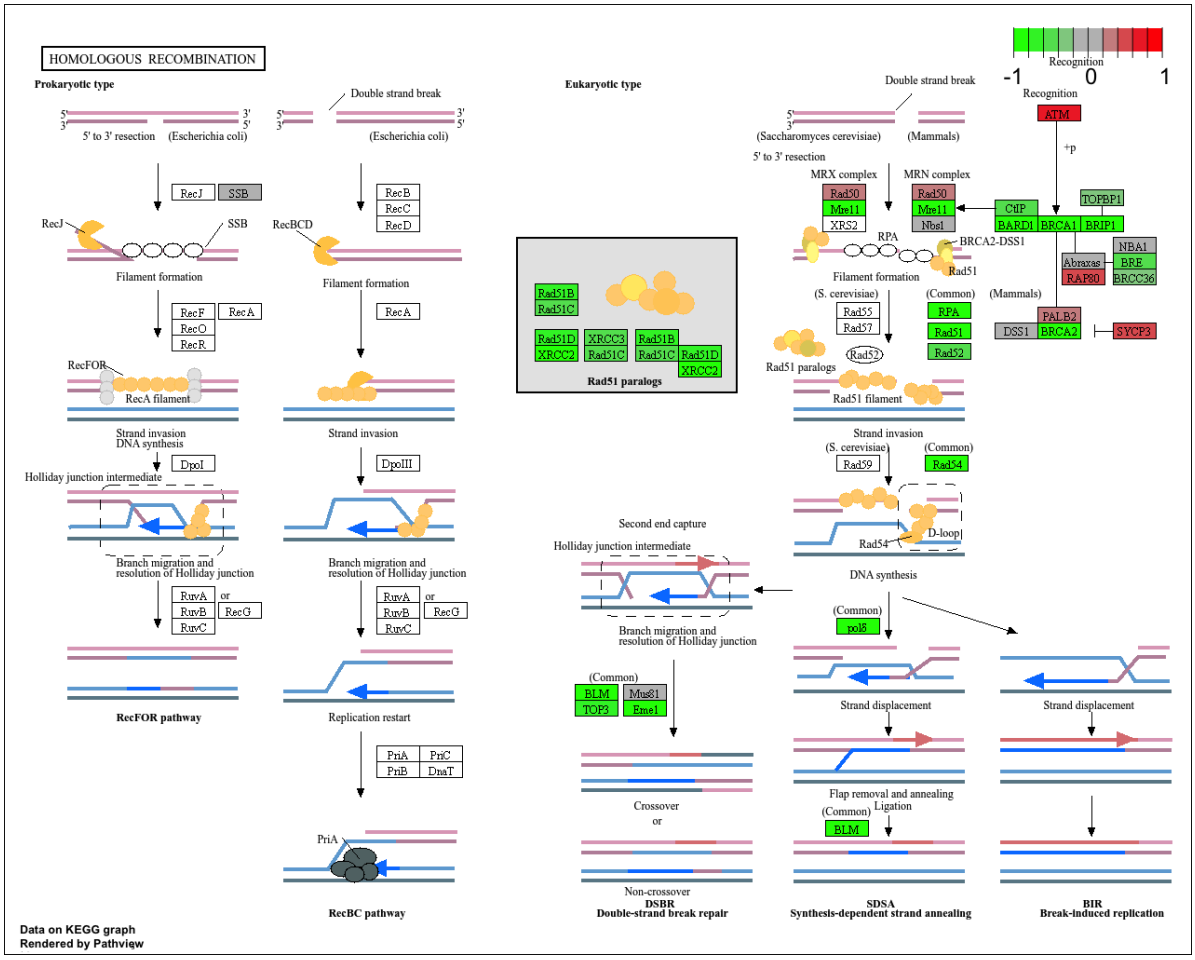
```
Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14

Info: Writing image file hsa04114.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/ikish/Desktop/Class14

Info: Writing image file hsa03440.pathview.png
```


CELL CYCLE
-1
0
1
ATRX
DDX11
Smc1
Smc3
NIPBL
Rad21
MAU2
Stag1,2
Cohesin loading
HDAC8
ESCO1/2
Smc1
Smc3
Rad21
Stag1,2
PDS5
Wapl
Sororin
Cohesion establishment
Plk1
CDK1
AuroraB
Sgo1
PP2A
Smc1
Smc3
Rad21
Stag1,2
Cohesin
Espl
Separin
PTTG
Securin
Growth factor
Growth factor withdrawal
TGFβ
p107
E2F4,5
DP-1,2
Smad2,3
Smad4
GSK3β
c-Myc
Miz1
MAPK signaling pathway
SCF
Skp2
p16
Ink4a
p15
Ink4b
p18
Ink4c
p19
Ink4d
p27,57
Kip1, 2
p21
Cip1
ARF
Mdm2
Rb
p300
DNA-PK
ATM/ATR
DNA damage checkpoint
p53
Apoptosis
GADD45
14-3-3σ
Chk1, 2
PCNA
Cdc25A
Ndc80
Mps1
KNL1
Mad1
Mad2
BubR1
Bub1
Bub3
TRIP13
CMT2
APC/C
Cdc20
14-3-3
Cdc25B,C
Ubiquitin mediated proteolysis
R-point (START)
CycD
CDK4,6
CycE
CDK2
CycA
CDK2
CycH
CDK7
CycA
CDK1
CycB
CDK1
Plk1
SCF
Skp2
p107,130
Rb
Abl
HDAC
TICRR
MTBP
Cdc45
Rb
Wee
Myt1
APC/C
Cdh1
E2F4,5
DP-1,2
E2F1,2,3
DP-1,2
Cdc6
Cdt1
ORC
MCM
Emi1
Cdc14b
p21
Cip1
p53
Cdc7
Dbf4
DNA replication
DNA
S-phase proteins, CycE
DNA
DNA biosynthesis
G1
S
G2
M
Data on KEGG graph
Rendered by Pathview


DNA REPLICATION
-1
0
1
Replication complex (Bacteria)
Lagging strand
Leading strand
RNase H / Pol I
Removal of RNA primer
Gap-filling
DNA ligase
Joining of
Okazaki fragment
Pol III core
Clamp
γδ
complex
Primase
DnaG
Primer
DnaB
SSB
Helicase
DNA polymerase III holoenzyme
DNA polymerase III holoenzyme
Pol III core
γδ complex
Helicase
DnaB
Primase
DnaG
SSB
RNaseH
RNaseHI
RNaseHII
RNaseHIII
DNA polymerase I
DpoI
DNA ligase
Lig
Replication complex (Archaea)
Lagging strand
Leading strand
RNase H or Dna2
DNA ligase
DNA polymerase
Clamp
Clamp loader
RFC
Primase
MCM
RPA/SSB
Helicase
DNA polymerase B
PolB
DNA polymerase D
PolD1
PolD2
Helicase
MCM
Primase
Pri1
Pri2
RPA/SSB
RPA
Clamp
PCNA
Clamp loader
RfcS
RfcL
RNaseH
RNaseHI
RNaseHIII
Helicase
Dna2
Fen1
DNA ligase
Lig
Replication complex (Eukaryotes)
Lagging strand
Leading strand
RNase H or Dna2
DNA ligase I
DNA polymerase δ
complex
PCNA
RFC
DNA polymerase α-primase
complex
α-Prim
MCM
2-7
DNA polymerase ε
complex
RPA
Helicase
DNA polymerase α-primase complex
α1
α2
Pri1
Pri2
DNA polymerase δ complex
δ1
δ2
δ3
δ4
DNA polymerase ε complex
ε1
ε2
ε3
ε4
MCM complex (helicase)
Mcm2
Mcm3
Mcm4
Mcm5
Mcm6
Mcm7
RPA
RFA1
RFA2/4
RFA3
Clamp
PCNA
Clamp loader
RFC1
RFC2/4
RFC3/5
RNaseHI
RNaseHI
RNaseHIII
RNaseHIII
Helicase
Dna2
Fen1
DNA ligase
Lig1
Data on KEGG graph
Rendered by Pathview


NUCLEOCYTOPLASMIC TRANSPORT
-1
0
1
Import
Importin
NLS
NPC
Cytoplasmic fibrils
Cytoplasm
Cytoplasmic ring
Central channel
Lumenal ring
Spoke complex
Lumen
Nucleoplasmic ring
Nucleus
Nuclear basket
NLS
Ran GTP
Importin
Export
Exportin
NES
Pi
Ran GDP
Rae1
DDX 19
Nup 98
Nup358 complex
Nup 214
Nup62 complex
Nup107-160 complex
ELYS
Nup 153
Tpr
Exportin
Ran GTP
NES
mRNA Export
EJC
PYM
AUG
PABP
AAAAA
Upf1
Upf2
Tap
Ref/ Aly
Cytoplasm
Lumen
NPC
Nucleus
mRNA surveilance pathway
SRm160
Pinin
EJC
p15
Upf3
Tap
TREX
Ref/Aly
CBC
m7G
AAAAA
Nuclear Pore complex (NPC)
Cytoplasmic fibrils
ALADIN
hCG1
Gle1
DDX19
Rae1
Nup98
Nup214
Nup88
Nup358 complex
RanBP2
RanGAP
UBC9
SUMO
Cytoplasmic ring / Nucleoplasmic ring (Symmetrical nups)
Nup160
Nup85
Sec13
Nup107
Nup133
Nup96
Nup145
Seh1
Nup43
Nup37
ELYS
Central channel
Nup62
Nup58/45
Nup54
Spoke complex
Nup205
Nup188
Nup155
Nup93
Nup53
Nup59
Lumenal ring
NDC1
gp210
pom121
pom152
pom34
pom33
Nuclear basket
Tpr
Nup50
Nup153
Senp2
Nup2
Nup1
Nup60
Nuclear transport complex
Importin
IPOA
IPOB
Adaptor proteins
SPN1
Exportin
XPO
Ran
eEF1A
PHAX
CBC
NMD3
Exon-junction complex (EJC)
EJC inner core
Y14
MAGOH
MLN51
EIF4A3
EJC outer shell
ACIN1
SAP18
RNPS1
Pinin
Ref/Aly
Transiently interacting factors
Upf1
Upf2
Upf3
Tap
p15
UAP56
SRm160
PYM
Transcription-export (TREX) complex
THO subcomplex
THOC1
THOC2
THOC5
THOC6
THOC7
TEX1
Data on KEGG graph
Rendered by Pathview


OOCYTE MEIOSIS
-1
0
1
INS
IGF1
Progesterone
Progesterone-mediated oocyte maturation
IGF1R
?
PGR
AR
AC
Xenopus oocyte
cAMP
PKA
Aurora-A
CPEB
MAPK signaling pathway
Early translation of maternal mRNAs
Mos
MEK1
c-MAPK
Rsk1/2
C3H-4
Ringo
Cdc2
Cdk2
MAPK1/2
PP1
Plk1
Plkk1
SCF
β-TrCP
Emi1
Myt1
14-3-3
Cdc25C
Ubiquitin mediated proteolysis
APC/C
Cdc20
MPF
CycB/B3
Cdc2
Degradation of CycB
Prevention of DNA replication
Bipolar spindle formation
Meiosis I progression
Meiosis I exit
Prophase I
GVBD
Meiosis I
Plk1
CPEB
Late translation of maternal mRNAs
Late-late translation of maternal mRNAs
SCF
β-TrCP
Emi2
Bub1
Mad1/2
CycE
Cdk2
APC/C
Cdc20
Securin
Separase
Arm cohesin
Rec8
STAG3
SMC1
SMC3
MPF
CycB/B4
Cdc2
Meiosis II arrest
Sgo
B56
PP2A
Degradation of CycB
Meiosis II resumption
Pericentromeric cohesin
Meiosis II exit
Interkinesis
Meiosis II
Fertilization
Plc-ζ
IP3
IP3R
Ca2+
Calm
CaN
CamKII
Plk1
PP2A
SCF
β-TrCP
Emi2
APC/C
Cdc20
Cdc25C
Delay to reentry into Meiosis
Data on KEGG graph
Rendered by Pathview



GO

We can also do a similar procedure with gene ontology. Similar to above, go.sets.hs has all GO terms. go.subs.hs is a named list containing indexes for the BP, CC, and MF ontologies. Let's focus on BP (a.k.a Biological Process) here.

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)
```

```
$greater
                                            p.geomean stat.mean        p.val
GO:0007156 homophilic cell adhesion        1.733613e-05  4.210950 1.733613e-
05
GO:0048729 tissue morphogenesis            5.399194e-05  3.888869 5.399194e-
05
GO:0002009 morphogenesis of an epithelium  5.719588e-05  3.879053 5.719588e-
05
GO:0030855 epithelial cell differentiation 2.051573e-04  3.555054 2.051573e-
04
GO:0060562 epithelial tube morphogenesis   2.924611e-04  3.458763 2.924611e-
04
GO:0048598 embryonic morphogenesis         2.954529e-04  3.446965 2.954529e-
04
                                                 q.val set.size         exp1
GO:0007156 homophilic cell adhesion        0.07579358      137 1.733613e-05
GO:0048729 tissue morphogenesis            0.08335347      483 5.399194e-05
GO:0002009 morphogenesis of an epithelium  0.08335347      382 5.719588e-05
GO:0030855 epithelial cell differentiation 0.16435924      299 2.051573e-04
GO:0060562 epithelial tube morphogenesis   0.16435924      289 2.924611e-04
GO:0048598 embryonic morphogenesis         0.16435924      498 2.954529e-04

$less
                                           p.geomean stat.mean        p.val
GO:0048285 organelle fission             6.657865e-16 -8.169820 6.657865e-16
GO:0000280 nuclear division              1.804970e-15 -8.050607 1.804970e-15
GO:0007067 mitosis                       1.804970e-15 -8.050607 1.804970e-15
GO:0000087 M phase of mitotic cell cycle 4.778014e-15 -7.914486 4.778014e-15
GO:0007059 chromosome segregation        1.084735e-11 -6.974104 1.084735e-11
GO:0051301 cell division                 8.753221e-11 -6.454861 8.753221e-11
                                                 q.val set.size         exp1
GO:0048285 organelle fission             2.630443e-12      386 6.657865e-16
GO:0000280 nuclear division              2.630443e-12      362 1.804970e-15
GO:0007067 mitosis                       2.630443e-12      362 1.804970e-15
GO:0000087 M phase of mitotic cell cycle 5.222369e-12      373 4.778014e-15
GO:0007059 chromosome segregation        9.484921e-09      146 1.084735e-11
GO:0051301 cell division                 6.378181e-08      479 8.753221e-11

$stats
                                           stat.mean     exp1
```

```
GO:0007156 homophilic cell adhesion         4.210950 4.210950
GO:0048729 tissue morphogenesis             3.888869 3.888869
GO:0002009 morphogenesis of an epithelium   3.879053 3.879053
GO:0030855 epithelial cell differentiation  3.555054 3.555054
GO:0060562 epithelial tube morphogenesis    3.458763 3.458763
GO:0048598 embryonic morphogenesis          3.446965 3.446965
```

Reactome

Let's now conduct over-representation enrichment analysis and pathway-topology analysis with Reactome using the previous list of significant genes generated from our differential expression results above.

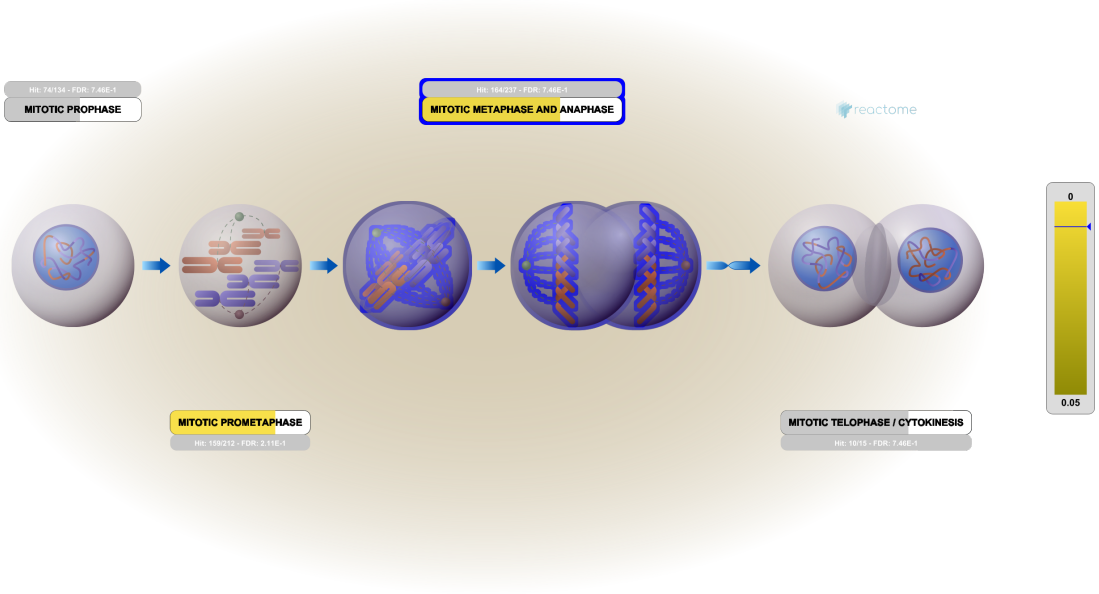
```r
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8156"
```

```r
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

> Q.What pathway has the most significant "Entities p-value"? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The pathway that has the most significant Entities p-value is the cell cycle, mitotic pathway which is consistent with the KEGG results since the cell cycle also had the most significant p-value in that results list.


Hit: 74/134 - FDR: 7.46E-1
MITOTIC PROPHASE
Hit: 164/237 - FDR: 7.46E-1
MITOTIC METAPHASE AND ANAPHASE
reactome
0
0.05
MITOTIC PROMETAPHASE
Hit: 159/212 - FDR: 2.11E-1
MITOTIC TELOPHASE / CYTOKINESIS
Hit: 10/15 - FDR: 7.46E-1